

STUDENT PERFORMANCE ANALYSIS USING PYTHON

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Domain: Data Analytics

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1. Objective

The objective of this project is to analyze student performance data using Python and data analysis libraries. The project focuses on understanding student academic performance, attendance trends, departmental performance, and identifying top-performing students through statistical analysis and data visualization.

2. Tools Used

The following tools and technologies were used in this project:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

3. Dataset Description

A sample dataset containing 50 student records was created and analyzed. The dataset consists of the following fields:

- Student ID
- Student Name
- Department
- Attendance Percentage
- Marks Obtained
- Semester

The dataset was stored in CSV format and loaded into Python using the Pandas library for further analysis.

A	B	C	D	E	F
Student ID	Student Name	Department	Attendance Percentage	Marks Obtained	Semester
1001	Student_1	MECH	88	54	3
1002	Student_2	CIVIL	80	78	2
1003	Student_3	ECE	82	50	3
1004	Student_4	CIVIL	95	79	8
1005	Student_5	ECE	81	92	2
1006	Student_6	MECH	89	77	2
1007	Student_7	MECH	80	72	4
1008	Student_8	IT	81	83	1
1009	Student_9	CSE	86	98	7
1010	Student_10	IT	87	99	8
1011	Student_11	ECE	62	76	3
1012	Student_12	CIVIL	68	78	2
1013	Student_13	MECH	84	99	6
1014	Student_14	IT	68	65	5
1015	Student_15	IT	79	67	7
1016	Student_16	MECH	66	83	5
1017	Student_17	ECE	73	56	4
1018	Student_18	IT	99	43	2
1019	Student_19	IT	63	93	5

4. Data Exploration

Data exploration was performed to understand the structure and quality of the dataset. The following operations were carried out:

- Displayed the first few records of the dataset.
- Examined dataset information and data types.
- Checked for missing values.
- Generated summary statistics for numerical columns.
- Verified data consistency and completeness.

The dataset did not contain missing values and was suitable for analysis.

```
print("First 5 Records")
print(df.head())

print("Dataset Information")
df.info()

print("Missing Values")
print(df.isnull().sum())

print("Summary Statistics")
print(df.describe())

-- First 5 Records
# Student ID Student Name Department Attendance Percentage Marks Obtained \
0 1001 Student_1 MECH 88 54
1 1002 Student_2 CIVIL 80 78
2 1003 Student_3 ECE 82 50
3 1004 Student_4 CIVIL 95 79
4 1005 Student_5 ECE 81 92

Semester
0 3
1 2
2 1
3 8
4 2

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 19 entries, 0 to 18
Data columns (total 6 columns):
# Column Non-Null Count Dtype
---
0 Student ID 19 non-null int64
1 Student Name 19 non-null object
2 Department 19 non-null object
3 Attendance Percentage 19 non-null int64
4 Marks Obtained 19 non-null int64
5 Semester 19 non-null int64
dtypes: int64(4), object(2)
memory usage: 2.1+ KB

Missing Values:
Student ID 0
Student Name 0
Department 0
Attendance Percentage 0
Marks Obtained 0
Semester 0
dtypes: int64

Summary Statistics:
# Student ID Attendance Percentage Marks Obtained Semester
count 19.000000 19.000000 19.000000 19.000000
min 1001.000000 62.000000 43.000000 1.000000
max 1019.000000 99.000000 99.000000 8.000000
std 16.577966 18.433333 27.455964 3.544166
qnt 1001.000000 66.000000 44.000000 1.000000
25% 1011.000000 72.000000 51.000000 2.000000
50% 1011.000000 81.000000 75.000000 4.000000
75% 1011.000000 86.000000 83.000000 5.000000
max 1019.000000 99.000000 99.000000 8.000000
```

5. Data Analysis

Several analytical operations were performed on the dataset:

Average Marks

The average marks of students were calculated to understand overall academic performance.

Highest and Lowest Marks

The maximum and minimum marks obtained by students were identified.

Department-wise Performance Analysis

Average marks were calculated for each department to compare academic performance across departments.

Attendance vs Marks Analysis

The relationship between attendance percentage and marks obtained was examined to determine whether attendance influences performance.

Top 10 Performing Students

The top 10 students were identified based on marks obtained.

```
# Average Marks
print("Average Marks:", df["Marks Obtained"].mean())

# Highest Marks
print("Highest Marks:", df["Marks Obtained"].max())

# Lowest Marks
print("Lowest Marks:", df["Marks Obtained"].min())

# Department Wise Performance
print("Department Wise Average Marks:")
print(df.groupby("Department")["Marks Obtained"].mean())

# Attendance vs Marks Correlation
print("Attendance vs Marks Correlation:")
print(df["Attendance Percentage"].corr(df["Marks Obtained"]))

# Top 10 Students
print("Top 10 Students:")
print(df.sort_values(by="Marks Obtained", ascending=False).head(10))
```

```
Average Marks: 71.96
Highest Marks: 99
Lowest Marks: 48

Department Wise Average Marks:
Department
CIVIL    77.142857
ECE      72.000000
IT       65.187500
MECH     68.000000

Name: Marks Obtained, dtype: float64

Attendance vs Marks Correlation:
-0.060338157757946

Top 10 Students:
  Student ID Student Name Department  Attendance Percentage  Marks Obtained \
12      1813  Student_13          MECH                      84                99
9       1818  Student_18          IT                       87                99
23      1824  Student_24          IT                       86                99
8        1809  Student_9           CSE                       86                98
28      1820  Student_20          ECE                       92                98
19      1810  Student_19          IT                       86                97
45      1846  Student_46          CSE                       68                96
25      1816  Student_16          CIVIL                      85                94
18      1819  Student_19          IT                       63                93
4        1805  Student_5           ECE                       81                92

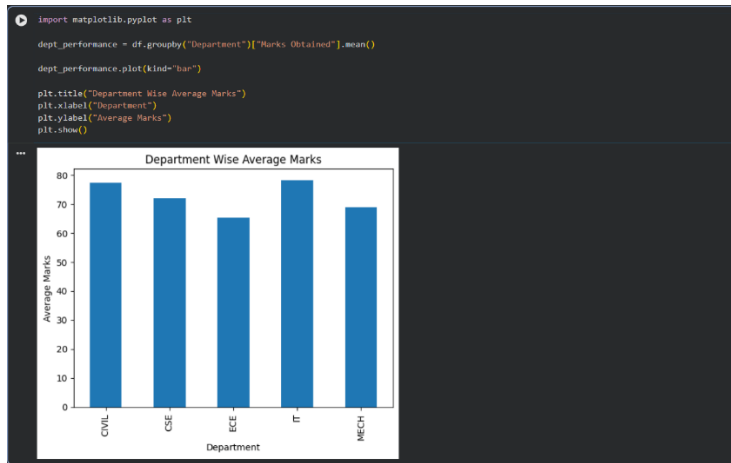
Semester
12      6
9       8
23      4
8       7
28      5
19      1
45      3
25      7
18      5
4       2
```

6. Data Visualization

Different charts were created to visualize the dataset and make the analysis easier to understand.

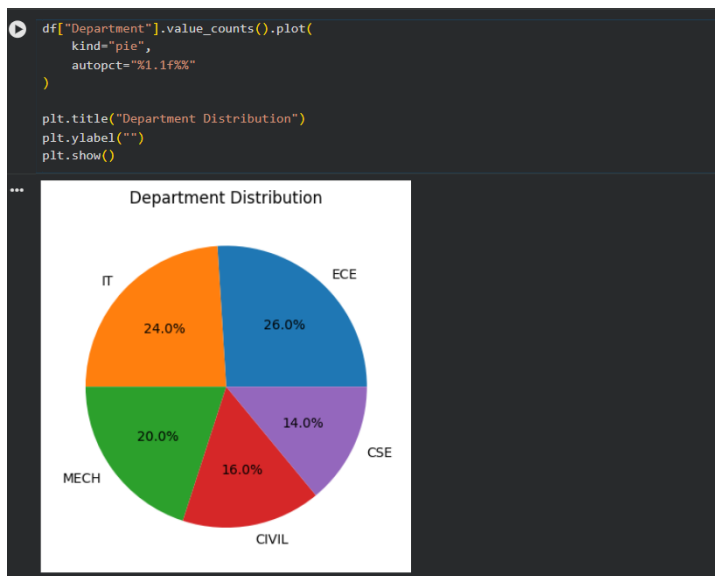
Bar Chart

A bar chart was created to display department-wise average marks.



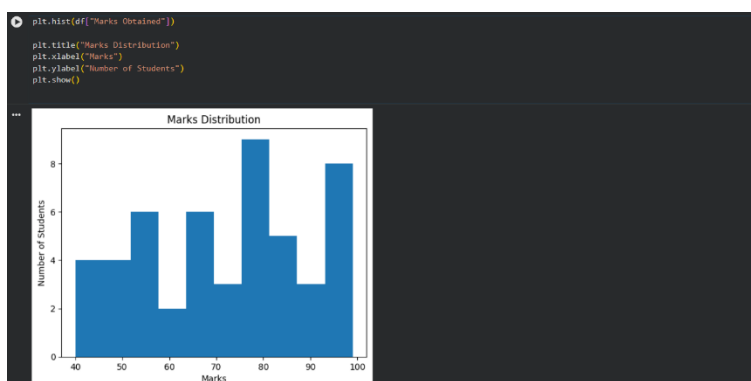
Pie Chart

A pie chart was used to show the distribution of students across departments.



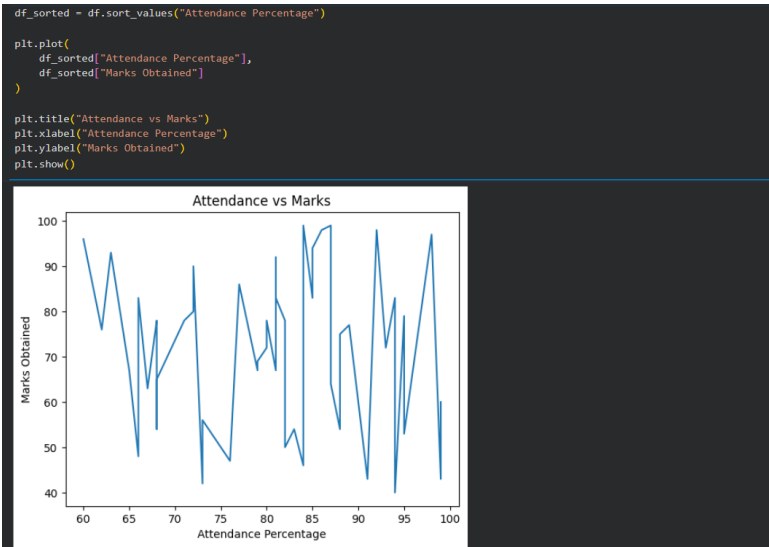
Histogram

A histogram was generated to analyze the distribution of marks among students.



Line Chart

A line chart was used to visualize the relationship between attendance percentage and marks obtained.



7. Key Insights

1. Student performance varied across different departments.
2. Students with higher attendance generally tended to achieve better marks.
3. The top-performing students scored significantly above the class average.
4. Most students scored within a moderate marks range as observed from the histogram.
5. Department-wise analysis helped identify differences in academic performance among departments.

8. Conclusion

The Student Performance Analysis project successfully demonstrated the use of Python for data analysis and visualization. By applying various analytical techniques, meaningful insights were obtained regarding student performance, attendance patterns, and departmental comparisons. The project highlights how data analytics can be used to support educational decision-making and evaluate academic outcomes effectively.